

정적크롤링 실습1. 명언 수집

- 명언을 제공하는 웹사이트에서 'life'관련 명언을 스크랩하여 엑셀 파일에 저장하기
- 대상 : <http://quotes.toscrape.com/>

크롤링하는 웹사이트 살펴보기

- 사이트에서 life 태그 클릭한 후 명언의 (about) 클릭
- 명언들 페이지의 하단 부분의 [Next]버튼, [Previous]버튼
- 살펴본 명언 웹사이트의 특징
 - 명언들이 태그별로 모여 있음
 - 한 페이지에 일정 수의 명언만 보이고 나머지는 다음(Next) 페이지를 통해 볼 수 있음

웹사이트 구조 확인 : 개발자도구(F12)

데이터 수집을 위한 절차

- 1단계 : 페이지 구조 확인 및 명언/저자가 있는 위치(태그, CSS선택자 등) 파악
- 2단계 : 웹페이지 불러오기(읽기)(url 요청 -> 응답)
- 3단계 : [각 태그(life)에 맞는] 명언/저자 추출 (BeautifulSoup 사용)

웹페이지 불러오기(읽기)

```
In [1]: # 모듈 импорт
import requests
from urllib.request import urlopen
from bs4 import BeautifulSoup
```

```
In [2]: # Life 명언 페이지 HTML 받아오기
url = 'https://quotes.toscrape.com/tag/life/'
html = urlopen(url).read()
soup = BeautifulSoup(html, 'html.parser')
```

각 태그에 맞는 명언/저자 추출(파싱)

```
In [4]: # 명언과 저자가 있는 부분 선택하여 추출 : <div class='quote'> => div.quote
quotes = soup.find_all('div', {'class':'quote'})
print(f'명언개수: {len(quotes)}')
```

명언개수: 10

```
In [5]: quotes[0]
```

```
Out[5]: <div class="quote" itemscope="" itemtype="http://schema.org/CreativeWork">
  <span class="text" itemprop="text">“There are only two ways to live your life. One is as thou
gh nothing is a miracle. The other is as though everything is a miracle.”</span>
  <span>by <small class="author" itemprop="author">Albert Einstein</small>
  <a href="/author/Albert-Einstein">(about)</a>
</span>
<div class="tags">
  Tags:
  <meta class="keywords" content="inspirational,life,live,miracle,miracles" itempro
p="keywords"/>
  <a class="tag" href="/tag/inspirational/page/1/">inspirational</a>
  <a class="tag" href="/tag/life/page/1/">life</a>
  <a class="tag" href="/tag/live/page/1/">live</a>
  <a class="tag" href="/tag/miracle/page/1/">miracle</a>
  <a class="tag" href="/tag/miracles/page/1/">miracles</a>
</div>
</div>
```

```
In [8]: quotes[0].find('span', {'class':"text"}).text
```

```
Out[8]: "There are only two ways to live your life. One is as though nothing is a miracle. The other
is as though everything is a miracle."
```

```
In [12]: quotes[0].find('span', {'class':"text"}).get_text(strip=True)
```

```
Out[12]: "There are only two ways to live your life. One is as though nothing is a miracle. The other
is as though everything is a miracle."
```

```
In [11]: quotes[0].find('span', {'class':"text"}).get_text().strip('"').strip('\'')
```

```
Out[11]: 'There are only two ways to live your life. One is as though nothing is a miracle. The other
is as though everything is a miracle.'
```

```
In [15]: quotes[0].find('small', {'class':"author"}).text
```

```
Out[15]: 'Albert Einstein'
```

```
In [16]: quotes[0].find('small', {'class':"author"}).get_text(strip=True)
```

```
Out[16]: 'Albert Einstein'
```

```
In [17]: # 명언, 저자 리스트를 따로 저장
texts, authors = [], []
for quote in quotes:
    texts.append(quote.find('span', {'class':"text"}).get_text().strip('"').strip('\''))
    authors.append(quote.find('small', {'class':"author"}).get_text(strip=True))
```

```
In [19]: authors
```

```
Out[19]: ['Albert Einstein',
'André Gide',
'Marilyn Monroe',
'Douglas Adams',
'Mark Twain',
'Allen Saunders',
'Dr. Seuss',
'Albert Einstein',
'George Bernard Shaw',
'Ralph Waldo Emerson']
```

```
In [28]: texts = [quote.find('span', {'class':"text"}).get_text().strip('"').strip('\'') for quote in q
authors = [quote.find('small', {'class':"author"}).get_text(strip=True) for quote in quotes]
print(f'저자:\n{authors}\n\n명언:\n{texts}')
```

저자:

```
['Albert Einstein', 'André Gide', 'Marilyn Monroe', 'Douglas Adams', 'Mark Twain', 'Allen Saunders', 'Dr. Seuss', 'Albert Einstein', 'George Bernard Shaw', 'Ralph Waldo Emerson']
```

명언:

```
['There are only two ways to live your life. One is as though nothing is a miracle. The other is as though everything is a miracle.', 'It is better to be hated for what you are than to be loved for what you are not.', 'This life is what you make it. No matter what, you're going to mess up sometimes, it's a universal truth. But the good part is you get to decide how you're going to mess it up. Girls will be your friends - they'll act like it anyway. But just remember, some come, some go. The ones that stay with you through everything - they're your true best friends. Don't let go of them. Also remember, sisters make the best friends in the world. As for lovers, well, they'll come and go too. And baby, I hate to say it, most of them - actually pretty much all of them are going to break your heart, but you can't give up because if you give up, you'll never find your soulmate. You'll never find that half who makes you whole and that goes for everything. Just because you fail once, doesn't mean you're gonna fail at everything. Keep trying, hold on, and always, always, always believe in yourself, because if you don't, then who will, sweetie? So keep your head high, keep your chin up, and most importantly, keep smiling, because life's a beautiful thing and there's so much to smile about.', 'I may not have gone where I intended to go, but I think I have ended up where I needed to be.', 'Good friends, good books, and a sleepy conscience: this is the ideal life.', 'Life is what happens to us while we are making other plans.', 'Today you are You, that is truer than true. There is no one alive who is Youer than You.', 'Life is like riding a bicycle. To keep your balance, you must keep moving.', 'Life isn't about finding yourself. Life is about creating yourself.', 'Finish each day and be done with it. You have done what you could. Some blunders and absurdities no doubt crept in; forget them as soon as you can. Tomorrow is a new day. You shall begin it serenely and with too high a spirit to be encumbered with your old nonsense.']
```

```
In [24]: # 딕셔너리 저장
quote_dict = {'text':[], 'author':[]}
for quote in quotes:
    quote_dict['text'].append(quote.find('span', {'class':"text"}).get_text().strip('"').strip())
    quote_dict['author'].append(quote.find('small', {'class':"author"}).get_text(strip=True))
quote_dict
```

```

Out[24]: {'text': ['There are only two ways to live your life. One is as though nothing is a miracle.
The other is as though everything is a miracle.',
'It is better to be hated for what you are than to be loved for what you are not.',
'This life is what you make it. No matter what, you're going to mess up sometimes, it's a u
niversal truth. But the good part is you get to decide how you're going to mess it up. Girls
will be your friends - they'll act like it anyway. But just remember, some come, some go. The
ones that stay with you through everything - they're your true best friends. Don't let go of
them. Also remember, sisters make the best friends in the world. As for lovers, well, they'll
come and go too. And baby, I hate to say it, most of them - actually pretty much all of them
are going to break your heart, but you can't give up because if you give up, you'll never fin
d your soulmate. You'll never find that half who makes you whole and that goes for everythin
g. Just because you fail once, doesn't mean you're gonna fail at everything. Keep trying, hol
d on, and always, always, always believe in yourself, because if you don't, then who will, sw
eetie? So keep your head high, keep your chin up, and most importantly, keep smiling, because
life's a beautiful thing and there's so much to smile about."],
'I may not have gone where I intended to go, but I think I have ended up where I needed to
be.',
'Good friends, good books, and a sleepy conscience: this is the ideal life.',
'Life is what happens to us while we are making other plans.',
'Today you are You, that is truer than true. There is no one alive who is Youer than You.',
'Life is like riding a bicycle. To keep your balance, you must keep moving.',
'Life isn't about finding yourself. Life is about creating yourself.',
'Finish each day and be done with it. You have done what you could. Some blunders and absur
dities no doubt crept in; forget them as soon as you can. Tomorrow is a new day. You shall be
gin it serenely and with too high a spirit to be encumbered with your old nonsense.'],
'author': ['Albert Einstein',
'André Gide',
'Marilyn Monroe',
'Douglas Adams',
'Mark Twain',
'Allen Saunders',
'Dr. Seuss',
'Albert Einstein',
'George Bernard Shaw',
'Ralph Waldo Emerson']}

```

```

In [31]: # (저자, 명언) 튜플을 리스트 요소로 저장
quote_list = []
for quote in quotes:
    text = quote.find('span', {'class':"text"}).get_text().strip('"').strip('\'')
    author = quote.find('small', {'class':"author"}).get_text(strip=True)
    quote_list.append((author, text))
quote_list

```

```
Out[31]: [('Albert Einstein',
          'There are only two ways to live your life. One is as though nothing is a miracle. The other is as though everything is a miracle. '),
          ('André Gide',
          'It is better to be hated for what you are than to be loved for what you are not. '),
          ('Marilyn Monroe',
          "This life is what you make it. No matter what, you're going to mess up sometimes, it's a universal truth. But the good part is you get to decide how you're going to mess it up. Girls will be your friends - they'll act like it anyway. But just remember, some come, some go. The ones that stay with you through everything - they're your true best friends. Don't let go of them. Also remember, sisters make the best friends in the world. As for lovers, well, they'll come and go too. And baby, I hate to say it, most of them - actually pretty much all of them are going to break your heart, but you can't give up because if you give up, you'll never find your soulmate. You'll never find that half who makes you whole and that goes for everything. Just because you fail once, doesn't mean you're gonna fail at everything. Keep trying, hold on, and always, always, always believe in yourself, because if you don't, then who will, sweetie? So keep your head high, keep your chin up, and most importantly, keep smiling, because life's a beautiful thing and there's so much to smile about. "),
          ('Douglas Adams',
          'I may not have gone where I intended to go, but I think I have ended up where I needed to be. '),
          ('Mark Twain',
          'Good friends, good books, and a sleepy conscience: this is the ideal life. '),
          ('Allen Saunders',
          'Life is what happens to us while we are making other plans. '),
          ('Dr. Seuss',
          'Today you are You, that is truer than true. There is no one alive who is Youer than You. '),
          ('Albert Einstein',
          'Life is like riding a bicycle. To keep your balance, you must keep moving. '),
          ('George Bernard Shaw',
          "Life isn't about finding yourself. Life is about creating yourself. "),
          ('Ralph Waldo Emerson',
          'Finish each day and be done with it. You have done what you could. Some blunders and absurdities no doubt crept in; forget them as soon as you can. Tomorrow is a new day. You shall begin it serenely and with too high a spirit to be encumbered with your old nonsense. ')]
```

명언과 저자를 추출하여 데이터프레임으로 저장하기

```
In [29]: # pandas 설치 및 импорт
!pip install numpy pandas
```

```
----- 1/2 [pandas]
----- 1/3 [pandas]
```

[illegible]

Successfully installed numpy-2.5.2 pandas-3.0.5

```
In [30]: import pandas as pd
pd.__version__
```

```
Out[30]: '3.0.5'
```

```
In [32]: df = pd.DataFrame(data=quote_list, columns=['저자', '명언'])
df
```

Out[32]:

	저자	명언
0	Albert Einstein	There are only two ways to live your life. One...
1	André Gide	It is better to be hated for what you are than...
2	Marilyn Monroe	This life is what you make it. No matter what,...
3	Douglas Adams	I may not have gone where I intended to go, bu...
4	Mark Twain	Good friends, good books, and a sleepy conscie...
5	Allen Saunders	Life is what happens to us while we are making...
6	Dr. Seuss	Today you are You, that is truer than true. Th...
7	Albert Einstein	Life is like riding a bicycle. To keep your ba...
8	George Bernard Shaw	Life isn't about finding yourself. Life is abo...
9	Ralph Waldo Emerson	Finish each day and be done with it. You have ...

In [33]:

```
df = pd.DataFrame(data=quote_dict)
df
```

Out[33]:

	text	author
0	There are only two ways to live your life. One...	Albert Einstein
1	It is better to be hated for what you are than...	André Gide
2	This life is what you make it. No matter what,...	Marilyn Monroe
3	I may not have gone where I intended to go, bu...	Douglas Adams
4	Good friends, good books, and a sleepy conscie...	Mark Twain
5	Life is what happens to us while we are making...	Allen Saunders
6	Today you are You, that is truer than true. Th...	Dr. Seuss
7	Life is like riding a bicycle. To keep your ba...	Albert Einstein
8	Life isn't about finding yourself. Life is abo...	George Bernard Shaw
9	Finish each day and be done with it. You have ...	Ralph Waldo Emerson

In [34]:

```
quotes = soup.select('div.quote')
quotes[0]
```

Out[34]:

```
<div class="quote" itemscope="" itemtype="http://schema.org/CreativeWork">
<span class="text" itemprop="text">“There are only two ways to live your life. One is as thou
gh nothing is a miracle. The other is as though everything is a miracle.”</span>
<span>by <small class="author" itemprop="author">Albert Einstein</small>
<a href="/author/Albert-Einstein">(about)</a>
</span>
<div class="tags">
  Tags:
  <meta class="keywords" content="inspirational,life,live,miracle,miracles" itempro
p="keywords"/>
  <a class="tag" href="/tag/inspirational/page/1/">inspirational</a>
  <a class="tag" href="/tag/life/page/1/">life</a>
  <a class="tag" href="/tag/live/page/1/">live</a>
  <a class="tag" href="/tag/miracle/page/1/">miracle</a>
  <a class="tag" href="/tag/miracles/page/1/">miracles</a>
</div>
</div>
```

```
In [35]: quotes[0].select_one('small.author').get_text(strip=True)
```

```
Out[35]: 'Albert Einstein'
```

```
In [37]: quotes[0].select('a.tag')
```

```
Out[37]: [<a class="tag" href="/tag/inspirational/page/1/">inspirational</a>,
<a class="tag" href="/tag/life/page/1/">life</a>,
<a class="tag" href="/tag/live/page/1/">live</a>,
<a class="tag" href="/tag/miracle/page/1/">miracle</a>,
<a class="tag" href="/tag/miracles/page/1/">miracles</a>]
```

```
In [44]: '|'.join([tag.get_text(strip=True) for tag in quotes[0].select('a.tag')])
```

```
Out[44]: 'inspirational|life|live|miracle|miracles'
```

```
In [45]: quotes[0].select('div.tags')[0]
```

```
Out[45]: <div class="tags">
      Tags:
      <meta class="keywords" content="inspirational,life,live,miracle,miracles" itemprop="keywords"/>
<a class="tag" href="/tag/inspirational/page/1/">inspirational</a>
<a class="tag" href="/tag/life/page/1/">life</a>
<a class="tag" href="/tag/live/page/1/">live</a>
<a class="tag" href="/tag/miracle/page/1/">miracle</a>
<a class="tag" href="/tag/miracles/page/1/">miracles</a>
</div>
```

```
In [50]: ' '.join(quotes[0].select('div.tags')[0].get_text().split()[1:])
```

```
Out[50]: 'inspirational life live miracle miracles'
```

문제. love 주제의 명언들 모두 추출하기

- 여러 페이지에 있는 명언들 모두 선택
- 명언과 저장와 태그명을 추출
- 데이터프레임으로 저장
- 함수로 작성해서 추출

```
In [51]: from urllib.request import urlopen
from bs4 import BeautifulSoup

tag_name = 'love'
page_no = 1
url = f'https://quotes.toscrape.com/tag/{tag_name}/page/{page_no}/'
html = urlopen(url).read()
soup = BeautifulSoup(html, 'html.parser')
# quotes = soup.find_all('div', {'class': 'quote'})
quotes = soup.select('div.quote')

quote_list = []
for quote in quotes:
    text = quote.select_one('span.text').get_text().strip('"').strip(' ')
    author = quote.select_one('small.author').get_text(strip=True)
    tags = '|'.join([tag.get_text() for tag in quote.select('a.tag')])
    # tags = ' '.join(quote.select('div.tags')[0].get_text().split()[1:])
    quote_list.append([author, text, tags])

df = pd.DataFrame(data=quote_list, columns=['저자', '명언', '태그'])
df
```

Out[51]:

	저자	명언	태그
0	André Gide	It is better to be hated for what you are than...	life love
1	Marilyn Monroe	This life is what you make it. No matter what,...	friends heartbreak inspirational life love sis...
2	Bob Marley	You may not be her first, her last, or her onl...	love
3	Elie Wiesel	The opposite of love is not hate, it's indiffe...	activism apathy hate indifference inspiraiona...
4	Friedrich Nietzsche	It is not a lack of love, but a lack of friend...	friendship lack-of-friendship lack-of-love lov...
5	Pablo Neruda	I love you without knowing how, or when, or fr...	love poetry
6	Marilyn Monroe	If you can make a woman laugh, you can make he...	girls love
7	Marilyn Monroe	The real lover is the man who can thrill you b...	love
8	James Baldwin	Love does not begin and end the way we seem to...	love
9	Jane Austen	There is nothing I would not do for those who ...	friendship love

In [52]:

```
def quote_extract(subject):
    page_no = 1
    quote_list = []

    while True:
        url = f'https://quotes.toscrape.com/tag/{subject}/page/{page_no}/'
        html = urlopen(url).read()
        soup = BeautifulSoup(html, 'html.parser')
        # quotes = soup.find_all('div', {'class': 'quote'})
        quotes = soup.select('div.quote') # 선택자 없으면 [] 반환

        if not quotes: # 더 이상 추출대상이 없는 경우
            break

        for quote in quotes:
            text = quote.select_one('span.text').get_text().strip('"').strip('\'')
            author = quote.select_one('small.author').get_text(strip=True)
            tags = ' '.join([tag.text for tag in quote.select('a.tag')])
            # tags = ' '.join(quote.select('div.tags')[0].get_text().split()[1:])
            quote_list.append([author, text, tags])
        page_no += 1

    return quote_list
```

In [53]:

```
subject = 'love'
quote_list = quote_extract(subject)
df = pd.DataFrame(data=quote_list, columns=['저자', '명언', '태그'])
df
```

Out[53]:

	저자	명언	태그
0	André Gide	It is better to be hated for what you are than...	life love
1	Marilyn Monroe	This life is what you make it. No matter what,...	friends heartbreak inspirational life love sis...
2	Bob Marley	You may not be her first, her last, or her onl...	love
3	Elie Wiesel	The opposite of love is not hate, it's indiffe...	activism apathy hate indifference inspira...
4	Friedrich Nietzsche	It is not a lack of love, but a lack of friend...	friendship lack-of-friendship lack-of-love lov...
5	Pablo Neruda	I love you without knowing how, or when, or fr...	love poetry
6	Marilyn Monroe	If you can make a woman laugh, you can make he...	girls love
7	Marilyn Monroe	The real lover is the man who can thrill you b...	love
8	James Baldwin	Love does not begin and end the way we seem to...	love
9	Jane Austen	There is nothing I would not do for those who ...	friendship love
10	C.S. Lewis	To love at all is to be vulnerable. Love anyth...	love
11	Alfred Tennyson	If I had a flower for every time I thought of ...	friendship love
12	Jane Austen	A lady's imagination is very rapid; it jumps f...	humor love romantic women
13	J.M. Barrie	To die will be an awfully big adventure.	adventure love

문제. 주제를 입력하면 해당 주제의 명언들을 추출하여 데이터프레임으로 저장

- 저자, 명언, 태그를 데이터프레임으로 저장

```
In [54]: subject = input('주제를 입력하세요! (love, life, humor, ...): ')
quote_list = quote_extract(subject)
df = pd.DataFrame(data=quote_list, columns=['저자', '명언', '태그'])
print(f'{subject}에 대한 명언들')
df
```

humor에 대한 명언들

Out[54]:

	저자	명언	태그
0	Jane Austen	The person, be it gentleman or lady, who has n...	aliteracy books classic humor
1	Steve Martin	A day without sunshine is like, you know, night.	humor obvious simile
2	Garrison Keillor	Anyone who thinks sitting in church can make y...	humor religion
3	Jim Henson	Beauty is in the eye of the beholder and it ma...	humor
4	Charles M. Schulz	All you need is love. But a little chocolate n...	chocolate food humor
5	Suzanne Collins	Remember, we're madly in love, so it's all rig...	humor
6	Charles Bukowski	Some people never go crazy. What truly horribl...	humor
7	Terry Pratchett	The trouble with having an open mind, of cours...	humor open-mind thinking
8	Dr. Seuss	Think left and think right and think low and t...	humor philosophy
9	George Carlin	The reason I talk to myself is because I'm the...	humor insanity lies lying self-indulgence truth
10	W.C. Fields	I am free of all prejudice. I hate everyone eq...	humor sinister
11	Jane Austen	A lady's imagination is very rapid; it jumps f...	humor love romantic women
